# ADVANCED DATA ANALYSIS

*For the Social Sciences · Graduate Course*

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**Format:** 14 weekly sessions · lecture + hands-on lab · Python (Anaconda / Google Colab / Jupyer / VS Code)
**Prerequisites:** No prior programming experience is assumed. Curiosity, patience, and a willingness to debug are.

## COURSE DESCRIPTION

Advanced Data Analysis is a demanding, hands-on course designed to bridge the social sciences and the computational tools that are reshaping how social research is done. It introduces programming fundamentals and a research-oriented computational toolkit, from the first line of Python to the design of language-model pipelines for social scientific measurement.

The course is organised in five movements. We begin with the fundamentals of Python and numerical computing, then move to data wrangling and the collection of digital trace data from the web. The third part covers machine learning as a research practice — model selection, evaluation, tuning — and closes by confronting prediction with causation, through the machine learning methods now used for causal estimation. The fourth part, which sets this course apart from a conventional data science curriculum, addresses the current frontier: using large language models as annotators, distilling their competence into small, transparent, reproducible classifiers, and building calibrated synthetic agents that stand in for human respondents. The fifth is an elective session whose content the class sets collectively, according to what the final projects turn out to require.

Throughout, technical instruction is paired with a curated set of readings that supply the theoretical and ethical backdrop. The aim is not to make you a machine learning engineer. It is to make you a social scientist who can decide, defensibly, which computational technique answers your question, and who can implement it, validate it, and report it.

### Learning Outcomes

By the end of the semester, students will be able to:

- write and debug idiomatic Python for data-intensive research tasks;
- acquire, clean, merge, and reshape research data from files, APIs, and web sources;
- design a supervised learning pipeline and evaluate it with metrics appropriate to the research question;
- distinguish predictive from causal questions, and apply machine learning methods built for causal estimation;
- use large language models as measurement instruments, with explicit validation against human-coded ground truth;
- distil a large model into a small, deployable classifier and justify the trade-off;
- reason critically about bias, representativeness, reproducibility, and research ethics in computational work;
- produce a complete, documented, publicly hosted research project in a GitHub repository.

## ASSESSMENT

| Component | Weight | Notes |
|---|---|---|
| Assignments (5) | 45% | Assignment 4 (machine learning shared task) is graded out of 13 points; the remaining four are graded out of 8 points each. |
| Midterm report | 5% | One page maximum. A research proposal in miniature. |
| Final project | 30% | Class presentation, written report (~2,000 words), the Python code, and a GitHub repository containing all of it. Submit the repository link only. |
| Attendance & participation | 20% | Includes one in-class paper presentation (under 20 minutes) per student. |

## Assignments

Detailed instructions for each assignment are posted on the public GitHub page of the course. You are expected to create your own GitHub repository for this course (name it Python_Course or something equally boring), and to add the instructor and the teaching assistant as collaborators. If you have never used GitHub, do not worry: it is easy to learn, and the internet is full of guides. Each week you will upload your work (Python scripts, written reports as PDF) to this repository with a descriptive title, before the deadline. The readability of your code counts towards your grade, so comment it where a reader would otherwise have to guess.

- **Assignment 1 —** Programming fundamentals.
- **Assignment 2 —** Data wrangling with Pandas.
- **Assignment 3 —** Web scraping and APIs.
- **Assignment 4 —** Machine learning. You are expected to build a full analytical pipeline: preprocessing, model selection, training and testing, hyperparameter tuning, and a set of reported outputs. This assignment is organised as a shared task; higher scores are rewarded. Details will be announced in class.
- **Assignment 5 —** LLM-based annotation. Annotate a small corpus with a language model, validate it against a human-coded gold standard.

## Midterm Report

The midterm report should not exceed one page. Define a research purpose, describe which methods might work, list the sources you may use, and explain how you plan to access your data. Think of it as a brief research proposal. Be precise.

## Final Project

The final version of your project is uploaded to your GitHub repository, and contains both code and writing. The written report should state the research purpose and hypothesis, describe the analytical model, report the findings, and cite data sources and previous publications. Formatting: Times New Roman, 12 pt, single-spaced, submitted as PDF rather than as an inconsistently rendered Word file.

The space of acceptable projects is deliberately wide. Choose a problem that genuinely interests you. You might generate data that does not currently exist using machine learning or language models; you might explain a phenomenon in your own research area; you might tune a regression model or build a synthetic sample and run an experiment inside it. In a similar course years ago I ran a spatial regression, constructing spatial lags to control for spatial autocorrelation and demonstrating the resulting bias in the OLS estimates. You may use the techniques covered in class or introduce a new approach, as long as it is implemented in Python. Your evaluation rests on the level of effort you invest, but make sure that effort is meaningful for your own future research and for the computational social science community.

Reading the weekly papers and book chapters is mandatory unless marked "optional." Each student presents one paper during the semester, in under ten minutes.

## CORE MATERIALS

- VanderPlas, Jake. 2016. *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media. Open access.
- Shaw, Zed A. 2017. *Learn Python 3 the Hard Way*. 1st ed. Addison-Wesley. Site.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.
- Jurafsky, Daniel, and James H. Martin. *Speech and Language Processing*, 3rd ed. draft — reference chapters on transformers and LLMs. Open access.
- The official Python documentation. docs.python.org/3.
- Weekly readings, online resources, and Python notebooks distributed through the course GitHub page.

## WEEKLY SCHEDULE

*Four modules, fourteen weeks. Technical readings prepare the lab; substantive readings prepare the discussion.*

**MODULE I · PROGRAMMING FOUNDATIONS (WEEKS 1–3)**

### Week 1. *Introduction: Computational Social Science and the Toolkit*

- Course logistics, expectations, and how the four modules fit together
- Setting up the environment: Anaconda, Google Colab, installation guidelines
- Git and GitHub for research: repositories, commits, and reproducibility
- **Reading:** Lazer, D., et al. (2009). "Computational Social Science." Science 323(5915): 721–723.
- **Reading:** Edelmann, A., Wolff, T., Montagne, D., & Bail, C. (2020). "Computational Social Science and Sociology." Annual Review of Sociology 46: 61–81.

### Week 2. *Fundamentals of Python Programming I*

- Data types, lists, sets, dictionaries, basic operations, conditionals
- A good starting point for beginners: W3Schools Python
- **Technical reading:** Shaw, Exercises 0–25.
- **Reading:** King, G. (2011). "Ensuring the Data-Rich Future of the Social Sciences." Science 331(6018): 719–721.

### Week 3. *Fundamentals of Python Programming II*

- Functions, loops, comprehensions, error handling, and writing code others can read
- **Technical reading:** Shaw, Exercises 26–40; or the official Python Tutorial.
- **Reading:** Lazer, D. (2015). "The Rise of the Social Algorithm." Science 348(6239): 1090–1091.

▸ **Assignment 1 released — Programming fundamentals**

**MODULE II · DATA: WRANGLING AND COLLECTION (WEEKS 4–6)**

### Week 4. *Numerical Computing and Visualisation: NumPy and Matplotlib*

- Arrays, vectorisation, broadcasting; why loops are the enemy
- From lists to datasets; the grammar of a scientific plot with Matplotlib
- **Technical reading:** VanderPlas, Chapters 2 and 4 (selected sections).
- **Reading:** Salganik, M. (2018). Bit by Bit, Chapter 2 ("Observing Behavior"). Open access.

### Week 5. *Data Manipulation with Pandas*

- Series and DataFrames; indexing and selection; operating on data; missing data
- Merging, joining, grouping, and reshaping — the full wrangling workflow in one session
- **Technical reading:** VanderPlas, Chapter 3 (complete). Reference: Pandas User Guide.

- **Reading:** Olteanu, A., Castillo, C., Diaz, F., & Kiciman, E. (2019). "Social Data: Biases, Methodological Pitfalls, and Ethical Boundaries." Frontiers in Big Data 2: 13.
- **Reading:** Sen, I., Flöck, F., Weller, K., Weiß, B., & Wagner, C. (2021). "A Total Error Framework for Digital Traces of Human Behavior on Online Platforms." Public Opinion Quarterly 85(S1): 399–422.

▸ **Assignment 2 released — Data wrangling with Pandas**

### Week 6. *Web Scraping and APIs*

- Techniques and tools for extracting web data: requests, Scrapy (Emlakjet)
- Working with APIs; rate limits, pagination, authentication (Youtube, BlueSky, GDELT)
- Legal and ethical boundaries of automated data collection; terms of service and the post-API era
- **Reading:** Jünger, J. (2018). "Mapping the Field of Automated Data Collection on the Web." In Stützer, Welker & Egger (eds.), Computational Social Science in the Age of Big Data. Köln: Halem Verlag, 104–130.
- **Reading:** Schäfer, M. T., & van Es, K. (2017). "Social Data APIs," Chapter 10 in The Datafied Society. Amsterdam University Press.
- **Reading:** Freelon, D. (2018). "Computational Research in the Post-API Age." Political Communication 35(4): 665–668.

▸ **Assignment 3 released — Web scraping and APIs** · ▸ **Midterm report due**

## MODULE III · MACHINE LEARNING FOR SOCIAL SCIENTIFIC RESEARCH (WEEKS 7–10)

### Week 7. *Machine Learning Essentials*

- What machine learning is, and what it is not: prediction versus explanation
- Supervised and unsupervised learning; the bias–variance trade-off; train/validation/test discipline
- First models with scikit-learn: classification, regression, clustering
- **Reading:** Molina, M., & Garip, F. (2019). "Machine Learning for Sociology." Annual Review of Sociology 45: 27–45.
- **Reading:** Hofman, J. M., et al. (2021). "Integrating Explanation and Prediction in Computational Social Science." Nature 595: 181–188.

### Week 8. *Model Selection, Evaluation, and Tuning*

- Cross-validation, hyperparameter search, class imbalance, and metric choice
- Algorithms in practice: k-nearest neighbours, penalised regression (ridge, lasso), tree ensembles
- **Reading:** Hindman, M. (2015). "Building Better Models: Prediction, Replication, and Machine Learning in the Social Sciences." ANNALS of the AAPSS 659(1): 48–62.
- **Reading:** Lones, M. A. (2021). "How to Avoid Machine Learning Pitfalls: A Guide for Academic Researchers." arXiv:2108.02497.
- **Technical reading:** VanderPlas, Chapter 5 (first half).

### Week 9. Advanced Machine Learning and *Neural Networks*

- Naive Bayes, SVM, random forests, PCA; then from perceptrons to deep architectures
- **Technical reading:** VanderPlas, Chapter 5 (second half); Jurafsky & Martin, Ch. 9–11.

▸ **Assignment 4 released — Machine learning shared task**

### Week 10. *Causal Inference and Machine Learning*

- Why a good predictor is not a causal estimate: selection, confounding, and the limits of fit
- Regularisation in the service of identification: post-double selection and double/debiased machine learning
- **Reading:** Angrist, J. D., & Pischke, J. S. (2009). “Chapter 2: Experimental Ideal” in the book titled “Mostly harmless econometrics: An empiricist's companion.” Princeton University Press.

- **Reading:** Brand, J. E., Zhou, X., & Xie, Y. (2023). "Recent Developments in Causal Inference and Machine Learning." Annual Review of Sociology 49: 81–110.

## MODULE IV · AI IN THE RESEARCH PIPELINE (WEEKS 11–12)

### Week 11. *LLMs as Annotators, and Distillation into a Deployable Model*

- Text as data: from bag-of-words and embeddings to the transformer architecture
- Fine-tuning a BERT-family encoder for a social-scientific classification task
- LLMs: Prompting as an operationalisation problem: task definition, codebooks, few-shot design, structured output
- Validation against human-coded gold standards
- Why distil (teacher-student relationship): cost, latency, reproducibility, data sovereignty, and the ability to re-run a frozen model five years from now
- **Reading:** Ziems, C., et al. (2024). "Can Large Language Models Transform Computational Social Science?" Computational Linguistics 50(1): 237–291.
- **Optional:** Törnberg, P. (2024). "Best Practices for Text Annotation with Large Language Models." arXiv:2402.05129.

▸ **Assignment 5 released — LLM annotation and distillation**

### Week 12. *Synthetic Agents / Digital Twins*

- Persona conditioning and algorithmic fidelity: can a model stand in for a respondent?
- Calibrating agents against survey microdata; distributional fidelity versus individual-level accuracy
- Where it breaks
- Generative agent architectures and experiments inside a synthetic public; the ethics of replacing people with plausible text
- **Reading:** Bisbee, J., et al. (2024). "Synthetic Replacements for Human Survey Data? The Perils of Large Language Models." Political Analysis 32(4): 401–416.
- **Optional:** Kına, M. F. & Ekmen, H. Y. (2026). Calibrating synthetic respondents: A within-persona experiment on public reactions to protest in Turkey [Preprint, under review at the Journal of Computational Social Science]. socArxiv: https://osf.io/preprints/socarxiv/hp7gz_v1

## MODULE V · ELECTIVE SESSION AND CLOSING (WEEKS 13–14)

### Week 13. *Elective Topic — Set by the Class*

*The content of this week is deliberately left open and will be fixed collectively around Week 8, once the midterm reports have been read and the shape of the final projects is clear. The purpose is to give one full session to whatever the class actually needs rather than to whatever the syllabus predicted in advance. Readings will be announced at least two weeks ahead.*

Depending on the projects on the table, plausible candidates include:

- unsupervised text analysis (topic models from LDA to BERTopic; embedding-based scaling and measurement);
- spatial analysis (spatial weight matrices, spatial dependence, spatial regression with GeoPandas and PySAL);
- a project workshop: code review, data problems, and collective debugging of work in progress.

## CLOSING

### Week 14. *Final Project Presentations*

- Ten-minute presentations and collective feedback
- **For your GitHub page:** Gandrud, C. (2013). "GitHub: A Tool for Social Data Set Development and Verification in the Cloud." SSRN 2199367.

- **Final project due — GitHub repository link**

## EXTENDED BIBLIOGRAPHY

*The following is a working reading list rather than a requirement. It is organised thematically and is intended to support final projects and dissertation work.*

### Additional Resources on Python


- Adhikari, A., DeNero, J., & Wagner, D. Computational and Inferential Thinking, 2nd ed. Open access.
- Bail, C. "Data Science & Society" course materials. dssoc.github.io.
- McKinney, W. (2022). Python for Data Analysis, 3rd ed. O'Reilly.
- Foster, I., Ghani, R., Jarmin, R. S., Kreuter, F., & Lane, J. (2020). Big Data and Social Science, 2nd ed. Chapman & Hall/CRC.
- Tunstall, L., von Werra, L., & Wolf, T. (2022). Natural Language Processing with Transformers. O'Reilly.
- Hugging Face NLP Course. huggingface.co/learn.
- Text analysis notebooks: python-for-text-analysis · NLP+CSS 201 tutorials.


### Large Language Models in Social Science: Annotation and Measurement


- Atreja, S., Ashkinaze, J., Li, L., Mendelsohn, J., & Hemphill, L. (2025). "Prompt Design Matters for Computational Social Science Tasks." ICWSM.
- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). "ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks." PNAS 120(30).
- Do, S., Ollion, É., & Shen, R. (2022). "The Augmented Social Scientist." Sociological Methods & Research.
- Heseltine, M., & Clemm von Hohenberg, B. (2024). "Large Language Models as a Substitute for Human Experts in Annotating Political Text." Research & Politics 11(1).
- Ollion, É., Shen, R., Macanovic, A., & Chatelain, A. (2024). "The Dangers of Using Proprietary LLMs for Research." Nature Machine Intelligence 6.
- Rathje, S., et al. (2024). "GPT Is an Effective Tool for Multilingual Psychological Text Analysis." PNAS 121(34).
- Törnberg, P. (2023). "ChatGPT-4 Outperforms Experts and Crowd Workers in Annotating Political Twitter Messages with Zero-Shot Learning." arXiv:2304.06588.
- Ziems, C., et al. (2024). "Can Large Language Models Transform Computational Social Science?" Computational Linguistics 50(1).


### Distillation, Fine-Tuning, and Efficient Models


- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL.
- Hinton, G., Vinyals, O., & Dean, J. (2015). "Distilling the Knowledge in a Neural Network." arXiv:1503.02531.
- Hsieh, C.-Y., et al. (2023). "Distilling Step-by-Step!" Findings of ACL.
- Hu, E. J., et al. (2022). "LoRA: Low-Rank Adaptation of Large Language Models." ICLR.
- Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). "DistilBERT." arXiv:1910.01108.
- Vaswani, A., et al. (2017). "Attention Is All You Need." NeurIPS.
- Wang, Y., et al. (2023). "Self-Instruct: Aligning Language Models with Self-Generated Instructions." ACL.
- West, P., et al. (2022). "Symbolic Knowledge Distillation: From General Language Models to Commonsense Models." NAACL.


### Synthetic Agents, Simulation, and Silicon Samples


- Aher, G., Arriaga, R. I., & Kalai, A. T. (2023). "Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies." ICML.
- Argyle, L. P., et al. (2023). "Out of One, Many: Using Language Models to Simulate Human Samples." Political Analysis 31(3).

- Bail, C. A. (2024). "Can Generative AI Improve Social Science?" PNAS 121(21).
- Bisbee, J., et al. (2024). "Synthetic Replacements for Human Survey Data? The Perils of Large Language Models." Political Analysis 32(4).
- Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). "Can AI Language Models Replace Human Participants?" Trends in Cognitive Sciences 27(7).
- Grossmann, I., et al. (2023). "AI and the Transformation of Social Science Research." Science 380(6650).
- Horton, J. J. (2023). "Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?" NBER Working Paper 31122.
- Park, J. S., et al. (2023). "Generative Agents: Interactive Simulacra of Human Behavior." UIST.
- Park, J. S., et al. (2024). "Generative Agent Simulations of 1,000 People." arXiv:2411.10109.
- Santurkar, S., et al. (2023). "Whose Opinions Do Language Models Reflect?" ICML.
- Törnberg, P., Valeeva, D., Uitermark, J., & Bail, C. (2023). "Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms." arXiv:2310.05984.


## Machine Learning and Causal Inference


- Athey, S., & Imbens, G. W. (2019). "Machine Learning Methods That Economists Should Know About." Annual Review of Economics 11: 685–725.
- Brand, J. E., Zhou, X., & Xie, Y. (2023). "Recent Developments in Causal Inference and Machine Learning." Annual Review of Sociology 49.
- Chernozhukov, V., et al. (2018). "Double/Debiased Machine Learning for Treatment and Structural Parameters." The Econometrics Journal 21(1).
- Chernozhukov, V., Newey, W. K., & Singh, R. (2022). "Automatic Debiased Machine Learning of Causal and Structural Effects." Econometrica 90(3).
- Egami, N., Fong, C. J., Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). "How to Make Causal Inferences Using Texts." Science Advances 8(42).
- Farbmacher, H., et al. (2022). "Causal Mediation Analysis with Double Machine Learning." The Econometrics Journal 25(2).
- Friedberg, R., Tibshirani, J., Athey, S., & Wager, S. (2021). "Local Linear Forests." Journal of Computational and Graphical Statistics 30(2).
- Li, F., Ding, P., & Mealli, F. (2023). "Bayesian Causal Inference: A Critical Review." Philosophical Transactions of the Royal Society A 381(2247).
- Ludwig, J., Mullainathan, S., & Rambachan, A. (2024). "Machine Learning as a Tool for Hypothesis Generation." Quarterly Journal of Economics 139(2).


## Big Data, Digital Traces, and the Epistemology of Computational Research


- Bail, C., et al. (2018). "Exposure to Opposing Views on Social Media Can Increase Political Polarization." PNAS 115(37).
- Boyd, D., & Crawford, K. (2012). "Critical Questions for Big Data." Information, Communication & Society 15(5).
- Freelon, D. (2018). "Computational Research in the Post-API Age." Political Communication 35(4).
- Golder, S. A., & Macy, M. W. (2014). "Digital Footprints: Opportunities and Challenges for Online Social Research." Annual Review of Sociology 40.
- Grimmer, J. (2015). "We Are All Social Scientists Now." PS: Political Science & Politics 48(1).
- Hargittai, E. (2015). "Is Bigger Always Better? Potential Biases of Big Data Derived from Social Network Sites." ANNALS of the AAPSS 659.
- Hofman, J. M., et al. (2021). "Integrating Explanation and Prediction in Computational Social Science." Nature 595.
- Jungherr, A., et al. (2017). "Digital Trace Data in the Study of Public Opinion." Social Science Computer Review 35(3).

- Kitchin, R. (2014). "Big Data, New Epistemologies and Paradigm Shifts." Big Data & Society 1(1).
- Lazer, D., & Radford, J. (2017). "Data ex Machina: Introduction to Big Data." Annual Review of Sociology 43.
- Lazer, D., et al. (2021). "Meaningful Measures of Human Society in the Twenty-First Century." Nature 595.
- Nelson, L. K. (2020). "Computational Grounded Theory." Sociological Methods & Research 49(1).
- Olteanu, A., Castillo, C., Diaz, F., & Kiciman, E. (2019). "Social Data: Biases, Methodological Pitfalls, and Ethical Boundaries." Frontiers in Big Data 2.
- Radford, J., & Lazer, D. (2019). "Big Data for Sociological Research." The Wiley Blackwell Companion to Sociology.
- Salganik, M. J. (2018). Bit by Bit: Social Research in the Digital Age. Princeton University Press.
- Salganik, M. J., et al. (2020). "Measuring the Predictability of Life Outcomes with a Scientific Mass Collaboration." PNAS 117(15).
- Sen, I., et al. (2021). "A Total Error Framework for Digital Traces of Human Behavior." Public Opinion Quarterly 85(S1).
- Wagner, C., et al. (2021). "Measuring Algorithmically Infused Societies." Nature 595.


## Bias, Ethics, and Accountability


- Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and Machine Learning. MIT Press. fairmlbook.org.
- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). "On the Dangers of Stochastic Parrots." FAccT.
- Birhane, A. (2021). "Algorithmic Injustice: A Relational Ethics Approach." Patterns 2(2).
- Bolukbasi, T., et al. (2016). "Man Is to Computer Programmer as Woman Is to Homemaker? Debiasing Word Embeddings." NeurIPS.
- Brayne, S. (2017). "Big Data Surveillance: The Case of Policing." American Sociological Review 82(5).
- Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). "Semantics Derived Automatically from Language Corpora Contain Human-Like Biases." Science 356(6334).
- Espeland, W., & Yung, V. (2019). "Ethical Dimensions of Quantification." Social Science Information 58(2).
- Hand, D. J. (2018). "Aspects of Data Ethics in a Changing World." Big Data 6(3).
- Taylor, L. (2017). "What Is Data Justice?" Big Data & Society 4(2).
- Weidinger, L., et al. (2022). "Taxonomy of Risks Posed by Language Models." FAccT.